



CANDIDATE – PLEASE NOTE!

PRINT your name on the line below and return this booklet with your answer sheet. Failure to do so may result in disqualification.

TEST CODE **01229010**

FORM TP 2018081

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**CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

INFORMATION TECHNOLOGY

Paper 01 – General Proficiency

1 hour 30 minutes

29 MAY 2018 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 30 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

Which of the following pairs represents general-purpose software tools?

- (A) Spreadsheet and database software
- (B) Word processor and accounting software
- (C) Student record system and database software
- (D) Insurance processing and spreadsheet software

Sample Answer



The best answer to this item is “Spreadsheet and database software”, so (A) has been shaded.

5. If you want to change your answer, erase it completely before you fill in your new choice.
6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. Which of the following items are considered computer hardware?

- (A) Data
- (B) Programs
- (C) Output devices
- (D) Operating systems

2. Scanning an item at a computerized checkout station in a supermarket is an example of data

- (A) input
- (B) output
- (C) storing
- (D) processing

3. Access time for magnetic disks is the time taken by the read/write head to find the

- (A) buffer
- (B) sector
- (C) cluster
- (D) cylinder

4. The binary equivalent of the decimal number 7 is

- (A) 100_2
- (B) 101_2
- (C) 110_2
- (D) 111_2

5. The primary storage device that holds the data currently being processed by the computer is known as the

- (A) read only memory (ROM)
- (B) programmable ROM (PROM)
- (C) random access memory (RAM)
- (D) erasable and programmable ROM (EPROM)

Item 6 refers to the following specification of a processor for a laptop.

Intel Pentium V 64-bit 3.6 GHZ

6. The 64-bit in this specification means that the processor can

- (A) process 4 characters at one time
- (B) process 8 characters at one time
- (C) access 8 characters at one time
- (D) access 4 characters at one time

7. Which of the following devices is used to input data into the computer?

- (A) Mouse
- (B) Monitor
- (C) USB flash drive
- (D) External hard drive

8. In which of the following interface types is the user required to remember specific instructions and syntax?

- (A) Menu-driven
- (B) Graphical user
- (C) Computer screen
- (D) Command-driven

9. Which of the following components is used by the CPU to temporarily store data?

- (A) Track
- (B) Sector
- (C) Buffer
- (D) Cylinder

10. In order to use a program stored on a flash drive, the program must be transferred into
- (A) ROM
 - (B) RAM
 - (C) PROM
 - (D) EPROM
11. Which of the following tasks is NOT a function of the control unit (CU)?
- (A) Performing arithmetic operations
 - (B) Reading and interpreting machine instructions
 - (C) Coordinating the operation of the internal registers
 - (D) Controlling the sequencing of machine instructions
12. The two's complement of a binary number is formed by
- (A) changing all zeros to twos
 - (B) changing all ones to twos
 - (C) finding the one's complement twice
 - (D) adding one to the one's complement
13. Which of the following terms represents a hexadecimal?
- (A) $5C_{16}$
 - (B) 10_2
 - (C) 15_{10}
 - (D) 23_8
14. A hard drive with **one** read/write head per disk is called a
- (A) SCSI
 - (B) SATA
 - (C) fixed head
 - (D) moving head
15. Which of the following systems provides transmission of data in BOTH directions at the same time?
- (A) Simplex
 - (B) Full duplex
 - (C) Half duplex
 - (D) Null duplex
16. Which of the following functions is performed by a modem?
- (A) Analog and digital
 - (B) Digital and upload
 - (C) Analog and download
 - (D) Download and upload
17. Which of the following devices is NOT required to successfully implement teleconferencing?
- (A) Sensor
 - (B) Modem
 - (C) Speaker
 - (D) Microphone
18. Programs that duplicate the knowledge and diagnostic skills displayed by humans performing specialized tasks are referred to as
- (A) virtual systems
 - (B) expert systems
 - (C) knowledge systems
 - (D) engineering systems
19. Transferring data from a remote computer to your computer is referred to as
- (A) browsing
 - (B) uploading
 - (C) data mining
 - (D) downloading

20. Which of the following operations is considered a feature of a biometric system?

- (A) Matching fingerprints
- (B) Detecting irregular heartbeats
- (C) Diagnosing medical symptoms
- (D) Helping to lubricate skeletal joints

21. Telecommuting can be BEST described as

- (A) using the computer while commuting
- (B) using the telephone while commuting
- (C) working from home by connecting home and office
- (D) working from the office after establishing a connection at home

22. A webmaster is important to a company because he/she

- (A) is responsible for the company's website
- (B) assigns web addresses to all users in a company
- (C) monitors the shutting down of competitors' websites
- (D) backs up files and stores the disks at another location

23. The task of keeping a company's data files and software organized through the cataloguing and storing of tapes and disks is that of the

- (A) IT manager
- (B) file librarian
- (C) systems analyst
- (D) database administrator

24. Which of the following checks is NOT a validation technique?

- (A) Range
- (B) Inconsistency
- (C) Double entry
- (D) Reasonableness

25. The process of scrambling messages at the sending end and unscrambling them at the receiving end is called

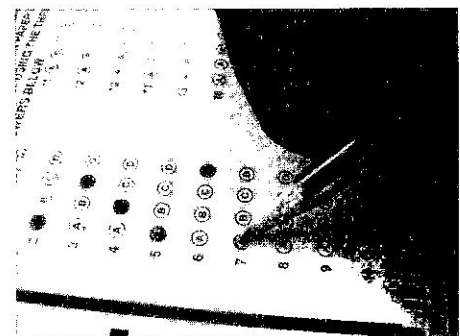
- (A) coding
- (B) encryption
- (C) espionage
- (D) eavesdropping

26. There are many software restrictions used by computers to protect data and maintain data integrity. Which of the following entities are software restriction media?

- I. Viruses
- II. Passwords
- III. Encryptions

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

Item 27 refers to the following diagram which shows a form being completed.



27. The data from the form shown above is captured by

- (A) USB
- (B) OCR
- (C) OMR
- (D) MICR

28. Which of the following statements is true of accessing files sequentially?
- (A) Occurs in any order
 - (B) Starts with the first record
 - (C) Starts with the largest data field
 - (D) Starts with the smallest data field
29. A graph is stored and sent to another user. When the user opens the file he understands the graph he receives. This graph is a form of
- (A) discussion
 - (B) facilitation
 - (C) calculation
 - (D) information
30. Which of the following activities is NOT an example of commercial data processing?
- (A) Budgeting
 - (B) Invoice processing
 - (C) Payroll processing
 - (D) Forecasting hurricanes
31. The following formula is located in position B6 of a spreadsheet. What would be the resulting formula if it was copied to the position located immediately on the left?
- $= B3 + B4 + \$B\5
- (A) $C3 + C4 + C5$
 - (B) $A3 + A4 + A5$
 - (C) $A3 + A4 + \$B\5
 - (D) $C3 + C4 + \$B\5
32. Spreadsheet formulae use
- (A) values or labels
 - (B) labels or cell addresses
 - (C) values, labels or cell addresses
 - (D) values, ranges or cell addresses
33. The intersection of a row and a column in a spreadsheet is called a
- (A) cell
 - (B) label
 - (C) range
 - (D) value
34. Which of the following would MOST likely be placed above a graph?
- (A) Title
 - (B) Legend
 - (C) Gridline
 - (D) Data label
35. In Microsoft PowerPoint, a single page shown on a screen to display information is called a
- (A) field
 - (B) table
 - (C) slide
 - (D) presentation
36. Which of the following is NOT a valid data type in Microsoft Access?
- (A) Text
 - (B) Character
 - (C) Date/time
 - (D) Auto number
37. Which of the following facilities is normally used for website-to-website connections?
- (A) Modem
 - (B) Monitor
 - (C) Hyperlink
 - (D) Hypertext

Item 38 refers to the following spreadsheet.

	A	B	C	D	E	F	G	H
1								
2								
3			Mon	Tue	Wed	Thur	Fri	Total
4		Games	195	657	200	100	700	=Sum(C4:G4)
5		Pool	290	410	311	150	390	=Sum(C5:G5)

38. If Column A is deleted, the existing formulae would
- (A) remain the same
 - (B) result in an error being reported
 - (C) be automatically corrected because absolute addressing is used
 - (D) be automatically corrected because relative addressing is used
-
39. Which of the following functions are performed by spreadsheets?
- I. Performing automatic re-calculation when values have been updated
 - II. Applying formulae to numerical data to determine values
 - III. Capturing, displaying and manipulating text data
- (A) I and II only
 - (B) I and III only
 - (C) II and III only
 - (D) I, II and III
40. In creating a query, a student entered the following criteria for the date field.
- >= 05/11/2017 and <= 05/14/2017
- The output will show days
- (A) after 14 May 2017 only
 - (B) from 11 to 14 May 2017
 - (C) before 11 May and after 14 May 2017
 - (D) between 14 May and 5 November 2017
41. Information produced from queries and tables in a specific format that can be printed from a database is called
- (A) a file
 - (B) a report
 - (C) a binary
 - (D) an object
42. The slide sorter can be used to
- (A) create a slide master
 - (B) modify custom animations
 - (C) add sound effects to a presentation
 - (D) show thumbnails of the entire slide show
43. Which of the following features can be used in a spreadsheet to copy the result across multiple rows in a single column?
- (A) Cell
 - (B) Label
 - (C) Cursor
 - (D) Function

44. In a spreadsheet, the column titles will always be visible when you scroll the spreadsheet if you first

- (A) freeze the top row
- (B) lock the last column
- (C) lock the spreadsheet
- (D) freeze the first column

45. Which of the following sequences of steps is correct for deleting a field in a database table?

- (A) Choose the delete row option.
Select row to be deleted and right click.
Open table in design view.
Choose yes to confirm deletion.
- (B) Open table in design view.
Select row to be deleted and right click.
Choose the delete row option.
Choose yes to confirm deletion.
- (C) Select row to be deleted and right click.
Open table in design view.
Choose the delete row option.
Choose yes to confirm deletion.
- (D) Select row to be deleted and right click.
Choose the delete row option.
Open table in design view.
Choose yes to confirm deletion.

46. Which of the following Pascal statements reads data into a variable, N, and places the cursor on the next line?

- (A) Read (N)
- (B) Write (N)
- (C) Readln (N)
- (D) Writeln (N)

Item 47 refers to the following pseudocode and trace table.

```

a ← 7
b ← 1
For n ← 1 to 3 do
  a ← b + 1
  b ← a + b
  
```

n	a	b
	7	1
1	2	X
2	4	7
3	8	15

47. What is the value of X?

- (A) 3
- (B) 8
- (C) 9
- (D) 14

48. The integer variables a, b and c are given the values 6, 10 and 2 respectively. What is the result of the following expression

a MULT (b DIV c)?

- (A) 0
- (B) 20
- (C) 30
- (D) 120

49. In representing algorithms, which of the following statements would be placed in the flow chart box shown below?



- (A) $A > 25$
- (B) Read A
- (C) Output C
- (D) $\text{Count} \leftarrow \text{count} + 1$

50. In an algorithm, `item_price` is a variable used to store the price of a grocery item. The MOST appropriate data type used for `item_price` is

(A) real
(B) integer
(C) boolean
(D) character

Item 51 refers to the following algorithm.

```
Read (A , B)
If A<B Then
Print(A);
```

51. Which of the following tasks does the following algorithm perform?

(A) Reads two numbers and prints the first number
(B) Reads two numbers and prints the smaller number
(C) Reads two numbers and prints the first number if it is larger than the second number
(D) Reads two numbers and prints the first number if it is smaller than the second number

52. In preparing a computer program, the programmer should FIRST

(A) plan the solution
(B) code the progress
(C) define the problem
(D) prepare program documentation

Item 53 refers to the following statements.

```
(1) PRINT "ENTER an integer"
(2) READ n
(3) While n <.> -1 DO
(4)     n = n - 1
(5)     Print n
(6) Endwhile
```

53. Which of the following series of numbers represents the correct output when the number 3 is input?

(A) 2, 1, 0
(B) 2, 1, 0, -1
(C) 2, 1, 0, -2
(D) 3, 1, 0, -1

54. The Pascal code `var x, y` would generate an error. The term used to describe this type of error is

(A) syntax
(B) logical
(C) run time
(D) execution

55. Which of the following terms is an invalid variable name?

(A) Sum
(B) 2count
(C) Vat15
(D) Count

56. The following commands are examples of computer programming instructions:

Sort customer by last name
Total income by department

To which generation of programming language do these examples belong?

(A) First
(B) Second
(C) Third
(D) Fourth

57. Which of the following symbols is used to perform assignments in the programming language Pascal?

- (A) =
- (B) :=
- (C) ←
- (D) ==

Item 58 refers to the following algorithm.

- I users for length, width
- II length, width
 $\text{area} \leftarrow \text{length} * \text{width}$
- III "Area of rectangle is", area

58. Which of the following options can be used to replace I, II and III in the algorithm?

- (A) Read, write, prompt
- (B) Write, read, prompt
- (C) Prompt, read, write
- (D) Read, prompt, write

59. Algorithm testing is BEST described as

- (A) executing the program manually with input values to see the results
- (B) running the program using sample data to identify errors in the program
- (C) reading each line of code from beginning to end to search for errors
- (D) checking that all documentation is included in the program

60. What is the MOST appropriate control structure to be used in calculating the total of ten numbers?

- (A) If-then-else
- (B) Repeat-until
- (C) While-do
- (D) For-do

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.